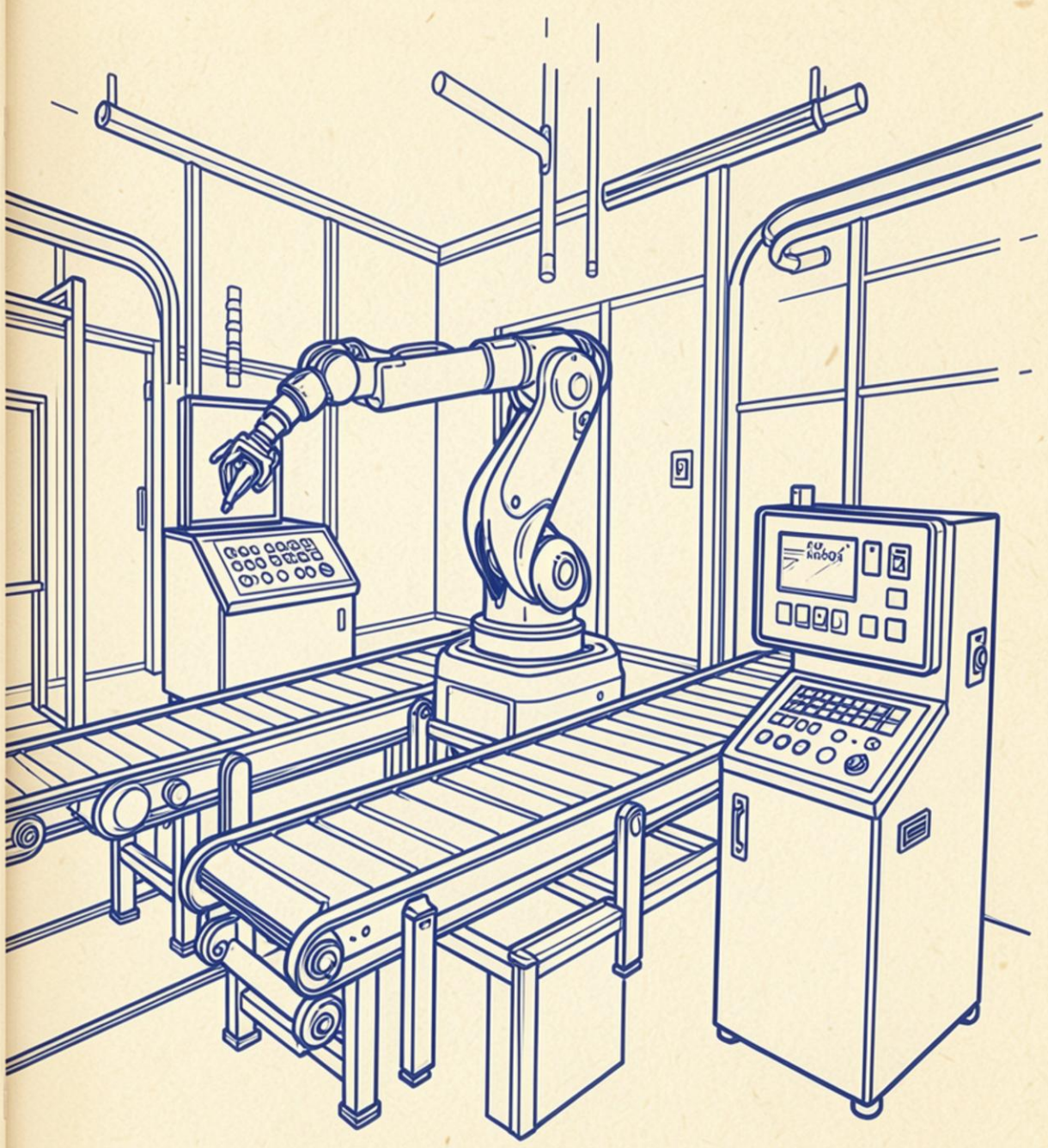
SMK MANUFACTURING ANALYTICS DASHBOARD

SMK Manufacturing Analytics Dashboard

End-to-End Manufacturing Analytics Solution

PREPARED BY: S.MANIKANDAN

Snowflake · SQL · Power BI · DAX · Dynamic RLS



Project Overview

The **SMK Manufacturing Analytics Dashboard** is an end-to-end **Business Intelligence solution** designed to monitor **Production, Quality, Downtime, Maintenance, and Machine Performance** across manufacturing plants.

GOAL

Enable data-driven decision making through centralized KPI monitoring and operational insights.

Business Problem

Fragmented reporting hides operational risk

- Lack of centralized reporting
- Production target tracking difficulties
- High defect rates
- Downtime monitoring issues
- Maintenance cost visibility challenges

SOLUTION

Power BI Analytics solution powered by Snowflake and secured with Dynamic RLS.



Project Objectives

The solution is built around six operating objectives



Production Monitoring



Quality Monitoring



Downtime Tracking



Maintenance Analysis



Machine Performance Analysis



Secure User Access

Deliver a centralized analytics platform for KPI monitoring and data-driven manufacturing decisions.

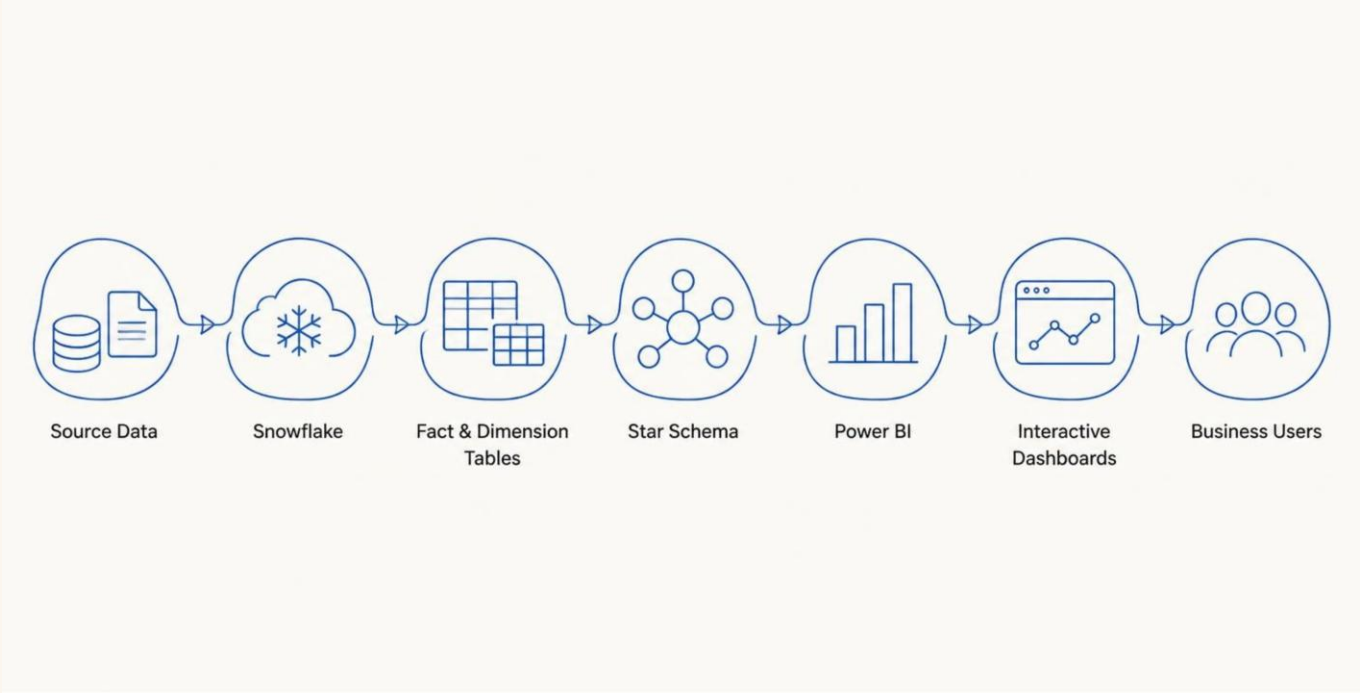
TECHNOLOGY STACK

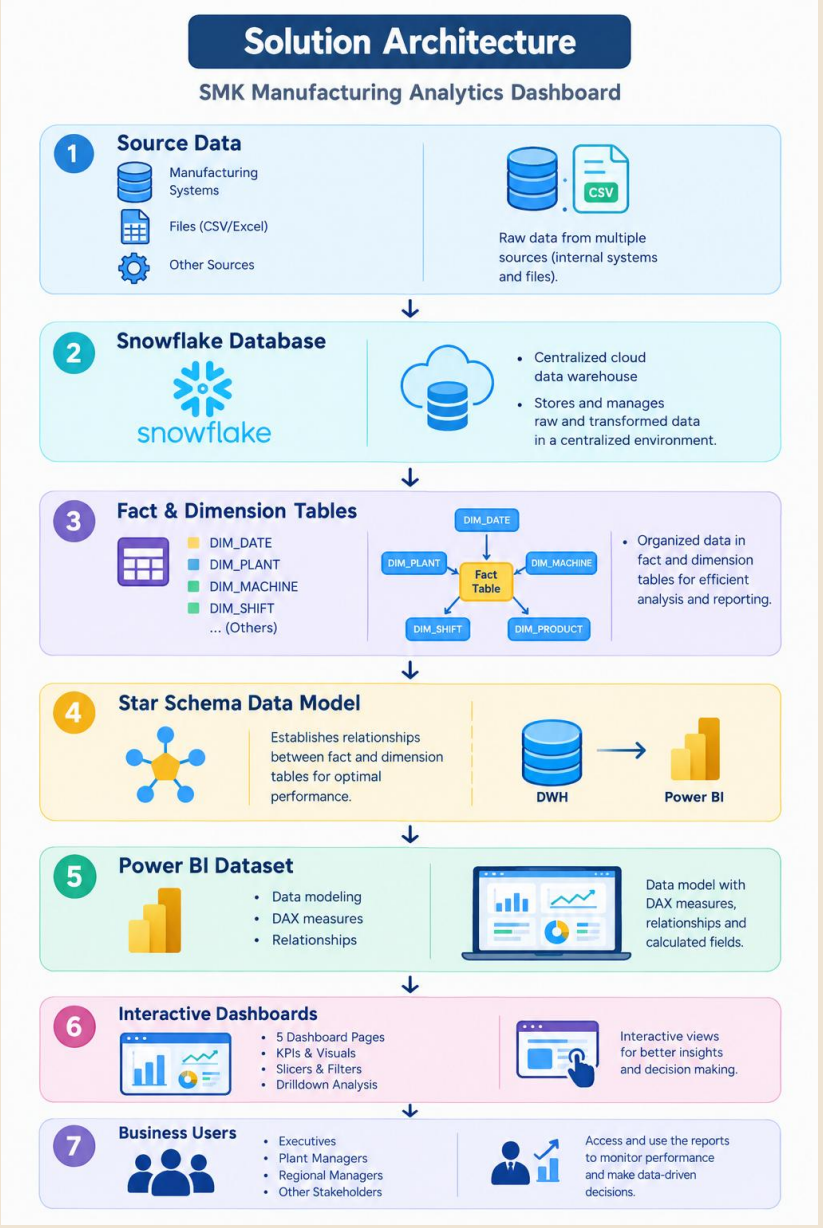
Each layer has a focused role

CATEGORY	TECHNOLOGY
Data Warehouse	Snowflake
Querying	SQL
Modeling	Star Schema
Visualization	Power BI Desktop & Service
Calculations	DAX
Security	Dynamic Row-Level Security (RLS)

ARCHITECTURE

From source data to business decisions





DATA MODEL OVERVIEW

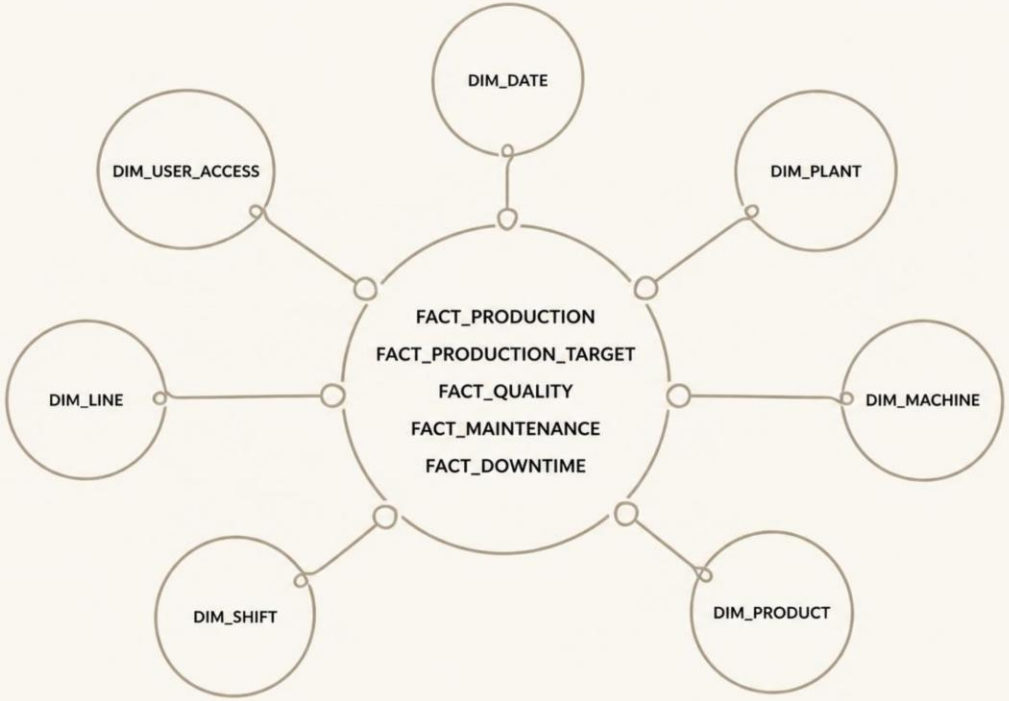
The star schema keeps analysis flexible

Fact tables:

- FACT_PRODUCTION
- FACT_PRODUCTION_TARGET
- FACT_QUALITY
- FACT_MAINTENANCE
- FACT_DOWNTIME

Dimension tables:

- DIM_DATE
- DIM_PLANT
- DIM_MACHINE
- DIM_PRODUCT
- DIM_SHIFT
- DIM_LINE
- DIM_USER_ACCESS



EXECUTIVE OVERVIEW

Executive overview puts the vital signals first

PRODUCTION ACHIEVEMENT %

95.42%

DEFECT RATE %

2.35%

MAINTENANCE COST

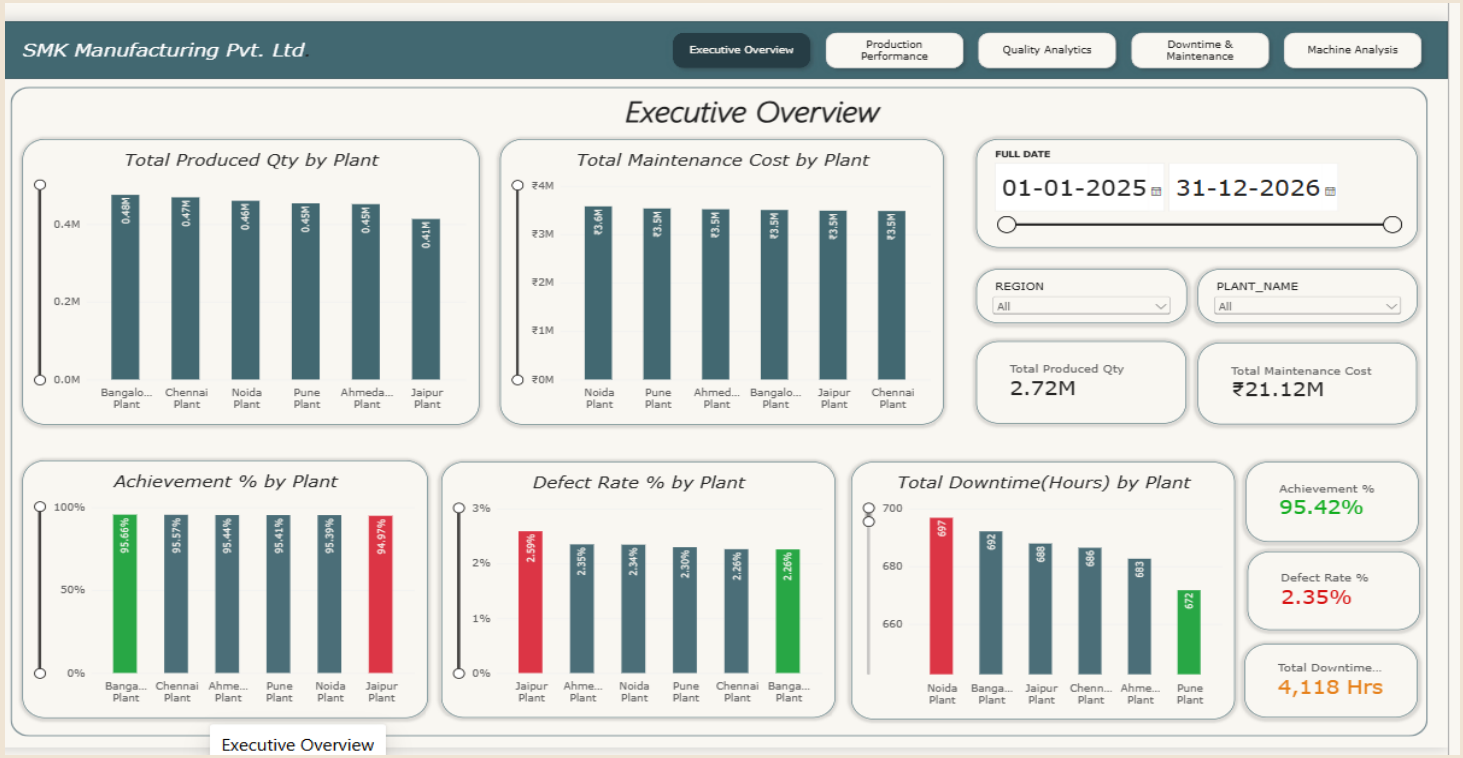
21.12 M

DOWNTIME HOURS

4,118 Hrs

Business value:

Provides a high-level manufacturing performance summary for executive decision-making.



- The Executive Overview dashboard provides a consolidated view of key manufacturing KPIs including Production Achievement, Defect Rate, Maintenance Cost, and Downtime Hours.
- It enables executives to monitor overall plant performance and quickly identify operational issues through a single reporting page.

PRODUCTION

Production performance connects plan to output

Total Produced Qty

2.72 M

Total Planned Qty

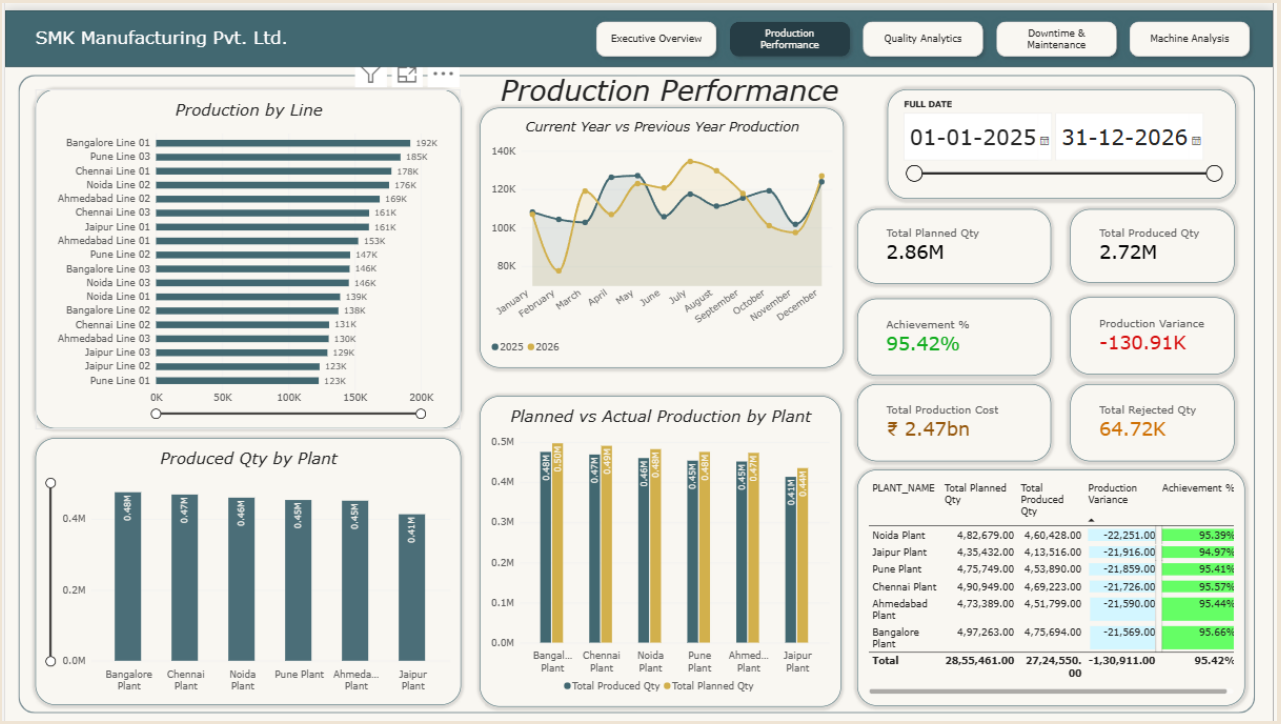
2.86 M

Production Variance

-130.91 K

Achievement %

95.42 %



Produced Qty by Plant

Drilldown: Plant → Line → Year → Quarter → Month

Planned Vs Actual Production by Plant

Drilldown: Plant → Year

- This dashboard analyzes production targets versus actual output using KPIs such as Production Quantity, Variance, and Achievement %, with drill-down from Plant to Month level.

QUALITY

Quality analytics makes defects actionable

TOTAL INSPECTIONS

2.68 M

PASS RATE %

97.65%

DEFECT RATE %

2.35%

TOTAL DEFECT QTY

62.97 K

Defects by Plant

Drilldown: Plant → Defect Type → Year



- This dashboard monitors inspections, pass rates, defect rates, and defect quantities.
- The drill-down functionality enables users to analyze defects from Plant to Defect Type and Year level.

Downtime & Maintenance analytics improves equipment performance

TOTAL MAINTENANCE COST

21.12 M

Cost per Event

6.52 K

Total Downtime Hours

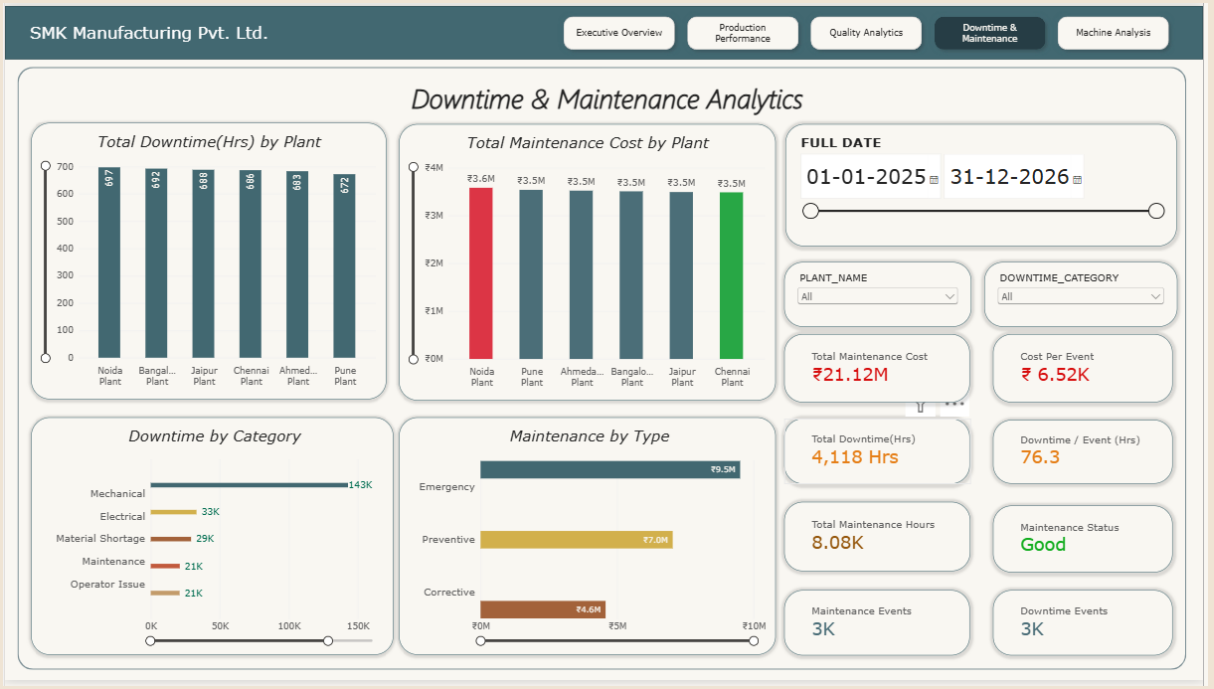
4,118 Hrs

Downtime Events

3k

Total Maintenance Cost by Plant
Drilldown: Plant → Machine Type → Year

Total Downtime(Hrs) by Plant
Drilldown: Plant → Year



- This dashboard analyzes maintenance costs, downtime hours, downtime events, and maintenance activities.
- It helps identify operational disruptions, optimize maintenance planning, and improve equipment reliability.

Machine analysis identifies performance leaders and bottlenecks

Best Machine

Grinding Machine

Worst Machine

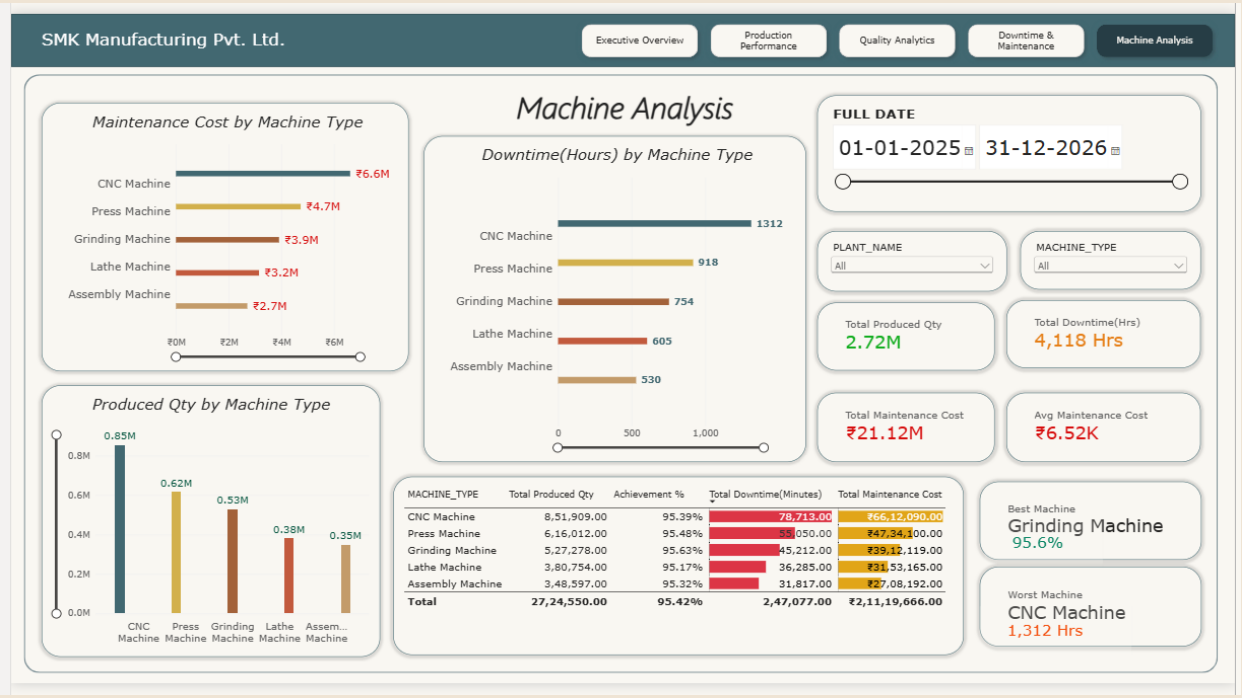
CNC Machine

TOTAL MAINTENANCE COST

21.12 M

Total Downtime Hours

4,118 Hrs



- This dashboard analyzes machine-level production, downtime, and maintenance performance across manufacturing plants.
- It helps identify the best-performing machines, track downtime trends, compare maintenance costs, and improve equipment utilization.

Dynamic RLS secures data by role
Role-Based Access Control

CEO
- Access to All Plants

Regional Manager
- Access to Assigned Region

Plant Manager
- Access to Assigned Plant

• Dynamic Row-Level Security (RLS) ensures that users can access only the data relevant to their assigned role.

• Security is implemented using USERPRINCIPALNAME() and DIM_USER_ACCESS to provide personalized dashboard access.

CEO View



Regional Manager View



Plant Manager View



Security Flow



Business Insights

Business insights transform data into decisions

PRODUCTION ACHIEVEMENT %

95.42%

Production Variance

-130.91 K

DEFECT RATE %

2.35%

TOTAL DEFECT QTY

62.97 K

TOTAL MAINTENANCE COST

21.12 M

Cost per Event

6.52 K

TOTAL DOWNTIME HOURS

4,118 Hrs

Total Downtime Events

3k

Best Machine

Grinding Machine

Worst Machine

CNC Machine

Key Business Insights

- Production operations achieved **95.42%** of the planned target, demonstrating strong overall manufacturing performance.
- The overall **Defect Rate of 2.35%** indicates effective quality control and inspection processes.
- Manufacturing plants recorded **4,118 Hours of Downtime**, highlighting opportunities for operational improvement.
- Total Maintenance Cost reached **₹21.12M**, enabling better maintenance planning and cost monitoring.
- **Grinding Machine** was identified as the best-performing machine, while **CNC Machine** recorded the highest downtime and requires focused maintenance attention.
- The dashboard provides centralized KPI monitoring and enables faster, data-driven decision-making across manufacturing operations.

Conclusion

From data to operational excellence

Project Outcome

- End-to-End Manufacturing Analytics Solution
- Snowflake Data Warehouse
- Star Schema Data Modeling
- Power BI Dashboard Development
- DAX KPI Calculations
- Dynamic Row-Level Security (RLS)
- Manufacturing Business Insights

Thank You

